Project Proposal Document

Project Title: Pregnancy Build Your Blood (PBYB) Challenge Web Platform

Deployment Model: Phased Release

Prepared By: Roverge Digital Lab

Date: October 10, 2026

Table of Contents

1. Project Overview & Objectives
2. Phase 1 Scope of Work (Week 1 Deployment)
3. Phase 2 Scope of Work (Core Challenge Engineering)
4. Technology Architecture
5. Phased Implementation Schedule
6. Budget & Payment Terms
7. Signatures
8. Appendix: AI Services for Green-Food Detection and PCV Result Reading

1. Project Overview & Objectives

The Pregnancy Build Your Blood (PBYB) Challenge platform is designed as an automated, mobile-first web ecosystem supporting a 1-week pre-challenge preparation phase followed by a 14-day main challenge cohort. To accommodate immediate marketing and pre-registration requirements, the project will be delivered in **two distinct phases**.

Phase 1 will establish the immediate public-facing infrastructure (onboarding, authentication, payments, and referral mechanics) within a rapid 1-week timeline. Phase 2 will subsequently deploy the complex Google Gemini Vision AI engines, gamification, educational checklists, and multi-channel communication hubs required for the main challenge.

2. Phase 1 Scope of Work (Week 1 Deployment)

Phase 1 focuses exclusively on securing user registrations, processing transactions, and incentivizing early sign-ups through affiliate gamification. This phase will be deployed to a live production environment within 7 days.

Authentication & Role-Based Onboarding

- **Secure Authentication:** Integration of standard Email/Password authentication alongside seamless Single Sign-On via **Google Auth** to minimize signup friction.
- **Dual-Role Profiling:** Dedicated onboarding flows distinguishing between **Pregnant Mums** (core participants providing gestational data) and **Nurses** (facilitators/support personnel), capturing distinct profile parameters and health consent waivers.

Automated Transaction Rails

- **Paystack Integration:** Exclusive integration of the Paystack payment gateway for processing the standard ₦5,000 registration fee.
- **Instant Provisioning:** Automated webhooks to instantly verify payments, issue receipts, and provision dashboard access upon successful checkout.

Referral Engine & Pre-Challenge Gamification

- **Affiliate Tracking:** Unique referral link and code generation for every registered user, tracking attributions for new sign-ups.
- **Referral Leaderboard:** A live, real-time leaderboard specifically ranking users by their number of successful referrals to drive viral pre-challenge registrations.

3. Phase 2 Scope of Work (Core Challenge Engineering)

Phase 2 introduces the artificial intelligence, cohort progression logic, and administrative workflows required to execute the 14-day challenge.

Gemini Vision AI Engine & Camera Integration

- **PCV Lab Document OCR:** Multimodal vision pipelines to read, parse, and validate baseline and completion PCV lab reports, extracting numerical values and testing dates into structured records.
- **Dietary Computer Vision:** AI evaluation of participant meals and drinks, verifying green dietary density before accepting logs.
- **Hardware Camera Enforcement:** Strict client-side constraints (capture="environment") to prevent device gallery uploads.

Dynamic Participant Engine & Educational Quizzes

- **Target Calibration:** Automatically segmenting participants into daily requirement tiers based on baseline PCV readings.
- **Cohort Progression & Checklists:** Enforcing calendar timelines for pre-challenge preparation and Day 1–14 unlocks.
- **Interactive Quizzes:** Multiple-choice quiz modules released by administrators with automated scoring.

Main Challenge Gamification & Administrative Payouts

- **Challenge Leaderboards:** Upload consistency tracking and an Overall Winner board tracking percentage PCV growth from start to completion.
- **Admin-Approved Payout Workflow:** Secure capture of affiliate bank details, commission balance ledgers (Available, Pending, Disbursed), and an admin console to approve payout batches and export CSV files for external bank disbursement.

Communications & Administrative CMS

- **Multi-Channel Dispatch:** Background workers for transactional messaging via WhatsApp, Email, SMS, and Telegram.
- **Admin CMS:** Central management for publishing daily lesson content (video/PDF/audio) and configuring prize distributions.

4. Technology Architecture

The platform will be built as a high-performance coupled monolith to maximize delivery velocity and accommodate the rapid Phase 1 deployment requirement:

- **Backend Framework:** Modern PHP (Laravel 12) or TypeScript (AdonisJS v6/v7) utilizing an MVC structure.
- **Frontend Layer:** Inertia.js coupled with React and Tailwind CSS, providing a responsive mobile-first SPA experience.
- **Database & Queue Management:** PostgreSQL for structured relational data and financial ledgers, alongside Redis for queue handling and cron tasks.
- **AI & Media Services:** Google Gemini Vision API; secure cloud storage with client-side image compression.

5. Phased Implementation Schedule

The project will be executed over a concentrated 5-week timeline, prioritizing the immediate rollout of registration tools.

Phase	Timeline	Key Deliverables
Phase 1: Registration Rollout	Week 1	• Database architecture and server provisioning • Google Auth and standard authentication • Pregnant Mum & Nurse onboarding workflows • Paystack gateway integration • Referral link generation and attribution • Public Referral Leaderboard **** Milestone: Live Production Deployment of Phase 1***
Phase 2: Vision AI & Progression	Week 2-3	• Gemini API integration (PCV OCR & dietary validation) • Hardware camera enforcement UI • Pre-challenge checklists and daily target allocation • Interactive quiz engine
Phase 2: Admin & Gamification	Week 4	• PCV growth and consistency leaderboards • Bank detail ledgers and admin payout CSV exports • Automated notification workers (Email/WhatsApp/SMS) • Admin CMS for content publishing
Phase 2: Hardening & Handover	Week 5	• End-to-end device QA (iOS Safari / Android Chrome) • Database indexing and AI vision fallback calibration • Final Phase 2 production deployment and handover

6. Budget & Payment Terms

The total investment for the complete ecosystem is **₦2,000,000**. Project financing follows a milestone-based distribution aligned with the phased delivery model:

- **Initial Commitment (50% - ₦1,000,000):** Due upon formal agreement execution prior to Phase 1 infrastructure provisioning.
- **Phase 1 Deployment Milestone (30% - ₦600,000):** Due upon the live production deployment of the Phase 1 registration, authentication, and referral systems at the end of Week 1.
- **Final Release & Handover (20% - ₦400,000):** Due upon user acceptance testing (UAT), Phase 2 production server deployment, and project handover at the end of Week 5.

For financial security, all our invoices will be sent via **finance@roverge.com**. Ensure the account details in the invoice match the ones below:

- **Account Holder:** Roverge Digital Lab
- **Bank Name:** Fidelity Bank
- **Account Number:** 5601575880

7. Signatures

Client Authorization	**Roverge Digital Lab Ltd.**
_____	_____
Name:	Nnebuchi Osigbo
Signature:	Chief Technology Officer
Date:	Date:

8. Appendix: AI Services for Green-Food Detection and PCV Result Reading

8.1 Purpose

The PBYB Challenge Platform requires an AI service capable of handling both of the following tasks:

- **Green-food and drink detection:** Analyse meal and drink photographs, identify visible food items or ingredients, and return a structured classification that the platform can use to calculate challenge scores.
- **PCV result extraction:** Read photographed laboratory reports, extract the PCV/haematocrit value and relevant details, and flag unreadable or ambiguous results.

The preferred solution should support image inputs, structured responses, asynchronous batch processing, and selective retries for failed or unclear images.

8.2 Estimated API Costs

The following estimates cover 2.6 million and 3 million image-analysis calls. Each call represents one image analysed by the selected AI model. These are planning estimates, not quotations. They assume approximately 1,182 input tokens per image, including the image and instructions, and 150 output tokens per response. Batch discounts are included where applicable. Actual image tokenisation, output length, retries, and model configuration will affect the final cost.

AI Service	Estimated cost for 2.6 million calls	Estimated cost for 3 million calls
Google Gemini 3.1 Flash-Lite (Recommended)	$677	$781
Anthropic Claude Haiku 5.5	$251	$290

AI Service	Estimated cost for 2.6 million calls	Estimated cost for 3 million calls
Mistral Small 4	$347	$401
OpenAI GPT-4.1 mini	$927	$1,069

Recommended option: Google Gemini 3.1 Flash-Lite

Gemini is recommended for the initial implementation because it offers multimodal image understanding, structured outputs, and batch processing within Google's AI ecosystem. The same model integration can be evaluated for both PCV extraction and green-food/drink analysis, potentially reducing the need to maintain separate AI providers. *Cost note:* Claude Haiku 5.5 has the lowest projected API cost; however, Gemini is the recommended starting point based on its overall fit for the proposed integration.

8.3 How Each Service Would Be Used

Service	Food and Drink Analysis	PCV Extraction	Batch Processing
Google Gemini 3.1 Flash-Lite	Analyse visible foods, drinks and green ingredients	Extract printed PCV values from report images	Yes
Anthropic Claude Haiku 5.5	Analyse food/drink images and return structured classifications	Extract PCV values and identify ambiguous reports	Yes
Mistral Small 4	Analyse image content and return structured results	Read report images and extract values	Yes
OpenAI GPT-4.1 mini	Analyse image content and return structured classifications	Extract PCV values from report images	Yes

All four options should be validated against the actual PBYB requirements before production use.

8.4 Batch Processing and Retry Strategy

The platform should assign every uploaded image a unique identifier and track its processing status independently of the batch containing it. Recommended workflow:

1. Store the image and its metadata.
2. Add the image to a batch-processing job.
3. Retrieve the result and match it to the original image ID.
4. Validate the returned fields and classify the result as completed, uncertain, or failed.
5. Retry only eligible failed or unclear images.
6. Send persistent failures or ambiguous PCV results for manual review or participant retake.

Batch processing is suitable for deferred scoring, but it should not be used for results that must be returned immediately after an upload.

8.5 Budget and Implementation Recommendation

Use Gemini 3.1 Flash-Lite as the first candidate for a controlled pilot, with Claude Haiku 5.5 as the cost-comparison alternative. Before approving a production budget:

- Test representative Nigerian meal and drink photographs.
- Test anonymised PCV laboratory reports with different layouts and image quality.
- Measure extraction accuracy, false classifications, retry rates, and actual token usage.

Final recommendation: Proceed with a Gemini pilot and compare it against Claude on the same test dataset. Confirm the final provider only after measuring accuracy and cost per successfully processed image.